Run report: cen-q5-cone-run-20261024T2208Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (sampled) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: cen-q5-cone 20261024T2208Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 cc0bcbe7b428feff4e2acbea3c6887d6abb5031b01f3a49589bd3597b3773dc7
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 457dc80ceded9acbbb0fb39ca1c61e8c3e2bfa773f863100584969c76131187d
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/cen-q5-cone-run-20261024T2208Z CENSUS 5 cone 50000000 202610245 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 136
```

tool_log.txt

```
cone 5: 43892004/50000000 samples, res-shaped 148504, allowed-trace cand 3620, smooth hits 3620, 120s
cone q=5: samples=50000000 (random) res_shaped=168992 allowed_trace_candidates=4128 smooth_prime_order_hits=4128 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1376/2000 secs=136.3
cone q=5: total prime-order mass estimate 41.992 +- 0.654 (hit rate per sampled form 8.256e-5)
```

summary.txt

```
cone q=5: samples=50000000 (random) res_shaped=168992 allowed_trace_candidates=4128 smooth_prime_order_hits=4128 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1376/2000 secs=136.3
cone q=5: total prime-order mass estimate 41.992 +- 0.654 (hit rate per sampled form 8.256e-5)
```

cand_t_hist.csv (first 12 of 85 rows)

type	t	candidates
cone	−50	522
cone	−49	231
cone	−46	1118
cone	−45	465
cone	−44	184
cone	−43	388
cone	−41	268
cone	−39	691
cone	−38	213
cone	−37	1431
cone	−36	91
cone	−35	109

hits.csv (first 12 of 4128 rows)

type	code	t	nE	smooth
cone	9985578608	−17	643	true
cone	4176399056	−21	647	true
cone	10209046869	−27	653	true
cone	27730488661	−33	659	true

cone	2615248492	−21	647	true
cone	17160923437	27	599	true
cone	3637424030	9	617	true
cone	12136877657	−33	659	true
cone	21965452193	−27	653	true
cone	15189766298	−17	643	true
cone	18032064696	27	599	true
cone	11154693818	25	601	true

mass_by_t.csv (first 10 of 10 rows)

type	t	nE	smooth_hits	mass_estimate	mass_stderr
cone	−33	659	683	6.9478	0.2659
cone	−27	653	964	9.8063	0.3158
cone	−21	647	406	4.1300	0.2050
cone	−17	643	604	6.1442	0.2500
cone	9	617	82	0.8341	0.0921
cone	13	613	398	4.0487	0.2029
cone	19	607	111	1.1292	0.1072
cone	25	601	285	2.8992	0.1717
cone	27	599	420	4.2725	0.2085
cone	39	587	175	1.7802	0.1346

Graph

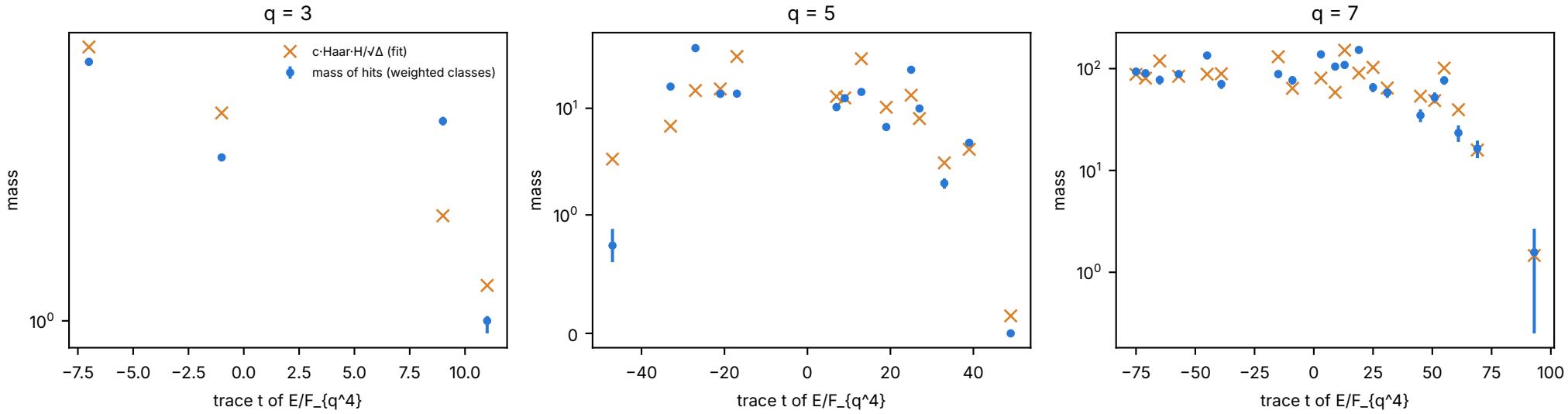


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.